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Structure and adsorption properties of commercial calcium phosphate adjuvant

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Abstract

Calcium phosphate adjuvant is a commercially available vaccine adjuvant that potentiates the immune response to antigens. Although its name suggests that it is $Ca_3(PO_4)_2$, X-ray diffraction, FTIR spectroscopy, thermal analysis and the Ca/P molar ratio identify commercial calcium phosphate adjuvant as non-stoichiometric hydroxyapatite, $Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$, where x varies from 0 to 2. The surface charge is pH-dependent (point of zero charge = 5.5). Consequently, commercial calcium phosphate adjuvant exhibits a negative surface charge at physiological pH and electrostatically adsorbs positively charged antigens. The presence of hydroxyls allows calcium phosphate adjuvant to adsorb phosphorylated antigens by ligand exchange with surface hydroxyls.
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Keywords: Commercial calcium phosphate adjuvant; Vaccine adjuvant; Non-stoichiometric hydroxyapatite; Point of zero charge; Electrostatic adsorption; Ligand exchange adsorption.

1. Introduction

Calcium phosphate adjuvant was developed at the Pasteur Institute and is used to potentiate the immune response of vaccines and to prepare adsorbed allergen extracts [1]. It is claimed to be a normal constituent of the body that is well tolerated and readily resorbed [1]. The World Health Organization and the European Pharmacopoeia recommend that it be used at concentrations of less than 1.3 mg of calcium per dose [1]. It is believed to potentiate the immune response by the depot mechanism whereby the antigen is adsorbed during the preparation of the vaccine and slowly released following administration. It is also believed to act by presenting the adsorbed antigen to antigen presenting cells as a particulate antigen [2].

A review of the literature found a report that calcium phosphate adjuvant has a molecular composition close to $Ca_3(PO_4)_2$ [3]. Another study found that the calcium/phosphorus molar ratio (Ca/P) varied from 1.35 to 1.83 depending on the rate of mixing during the precipitation reaction [2]. The properties of the precipitate were strongly dependent on the precipitation conditions. Calcium phosphate adjuvant precipitated by rapid mixing adsorbed 100% of diphtheria toxoid while the adjuvant produced by slow mixing only adsorbed 58% of the same dose of diphtheria toxoid. The incomplete and conflicting information regarding the composition and adsorption properties of calcium phosphate adjuvant stimulated the present study of its structure and adsorption properties.

2. Materials and methods

The calcium phosphate adjuvant used in this study was obtained from Reheis Inc. (Berkeley Heights, NJ). X-ray and FTIR analysis were also performed on calcium phosphate adjuvant obtained from Brenntag Biosector (Frederikssund,

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Denmark). All biochemicals and reagents were obtained from Sigma (St. Louis, MO) or Mallinckrodt (Paris, KY). The water was purified by a Barnstead Nanopure Infinity Ultrapure Water System (Dubuque, IA) and was 17 MΩ or better.

Samples for powder X-ray diffraction (XRD) were prepared by filtering a portion of the adjuvant and air drying the solid phase at room temperature. The powder was ground and prepared as a random powder mount in an aluminum holder. The powder XRD pattern was obtained on a Siemens D-500 diffractometer (Madison, WI) using CuKα radiation. Data were collected from 2 to 70° 2θ, counting for 2 s every 0.02° 2θ step.

The infrared spectrum from 4000 to 370 cm^{-1} was obtained with a Fourier-transform infrared spectrophotometer (model 2000 GX, Perkin-Elmer, Norwalk, CT). Potassium bromide pellets were prepared by mixing 1.5 mg of an air dried sample with 250 mg KBr.

Thermal analyses of the calcium phosphate adjuvant were obtained using differential scanning calorimetry (DSC) and thermal gravimetric analysis (TGA). Samples for DSC and TGA were prepared as described for XRD. DSC patterns of the calcium phosphate adjuvant were obtained using a Mettler DSC30 (Heightstown, NJ) equipped with a Mettler TC11 TA processor. Calcium phosphate adjuvant (7.1 mg) was placed in an aluminum pan and sealed using a crucible press. The DSC patterns were run from 30 to 575 °C at a rate of 10 °C min^{-1}. TGA patterns of the calcium phosphate adjuvant were obtained using a Dupont 951 TGA (Wilmington, DE). Calcium phosphate adjuvant (29.1 mg) was weighed in an alumina (Al_2O_3) pan in the sample chamber. The TGA patterns were run with a flow of air from 25 to 850 °C at a rate of 10 °C min^{-1}.

Photomicrographs were obtained using a transmission electron microscope (model 2000 FX, JEOL, Tokyo, Japan). The adjuvant was diluted with water until the solid content was 0.01%. The dilute suspension was sonicated for 15 min. A 300 mesh copper grid was dipped into the suspension and allowed to air dry.

The point of zero charge (PZC) was determined by the potentiometric titration technique initially described by Parks and de Bruyn [4]. This technique has been used to determine the PZC of hydroxyapatite [5]. The commercial calcium phosphate adjuvant containing 1.26% Ca was diluted with nanopure water to 1.00% Ca. A 10 mL sample was prepared by measuring 7.94 mL of the commercial adjuvant and adjusting the volume to 10 mL with nanopure water. The 10 mL sample was titrated by adding 10 μL increments of 0.1 M NaOH or HCl and recording the pH. The titration was repeated except 0.2 or 1.0 mL of 4 M KCl was added to the commercial adjuvant before the volume was adjusted to 10 mL. The volume of acid or base added did not exceed 2.39 or 1.58 mL, respectively.

Adsorption isotherms were prepared by mixing 2.5 mL of an adjuvant stock suspension containing 3.4 mg Ca/mL and 2.5 mL of the appropriate protein solution. Lysozyme solutions ranging from 0.2 to 3.2 mg/mL and alpha casein solutions ranging from 1.2 to 6.4 mg/mL were used. A preliminary experiment indicated that adsorption was complete in less than 1 h. Thus, the protein/adjuvant suspensions were mixed for 1 h. The samples were centrifuged and the supernatant assayed for protein by the bincinchoninic acid assay (Pierce, Rockford, IL). The amount of protein adsorbed was determined by the difference between the total protein and the protein in the supernatant.

3. Results and discussion

3.1. Structure

The crystalline structure of commercial calcium phosphate adjuvant was investigated using powder XRD (Fig. 1). The most intense features are reflections in the 31–34° 2θ range and a sharp, well-resolved feature at 25.9° 2θ. The strongest XRD reflection occurred at 31.8° 2θ. As seen in Table 1, the strongest powder XRD reflections of calcium and phosphate-containing phases also occur in the 31–34° 2θ region. However, the strongest XRD reflections of hydroxyapatite and calcium-deficient hydroxyapatite occur at 33.2° 2θ. The calcium phosphate adjuvant produced by Brenntag Biosector exhibited the same XRD pattern as shown in Fig. 1. The values corresponding to the atomic distances between layers (*d*-spacing) are also presented in Table 1.

Although the XRD patterns of hydroxyapatite and calcium-deficient hydroxyapatite are very similar, the calcium to phosphorus molar ratios (Ca/P) of hydroxyapatite and calcium-deficient hydroxyapatite are 1.67 and 1.50, respectively (Table 1). The Certificate of Analysis for the calcium phosphate adjuvant studied identified the Ca/P as 1.41. The Certificate of Analysis of a sample of calcium phosphate adjuvant produced by Brenntag Biosector, Elsenbakken, Denmark identified the Ca/P as 1.39.

Forms of hydroxyapatite known as non-stoichiometric hydroxyapatite [7] (NSHA) are used as bone substitutes [8] and

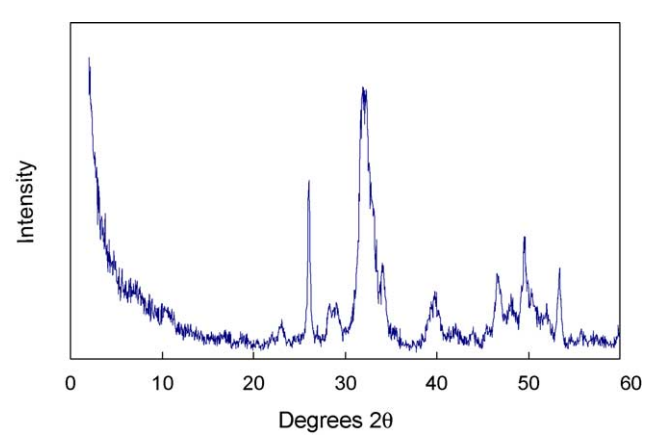

Fig. 1. Powder X-ray diffraction pattern of commercial calcium phosphate adjuvant.

Table 1
Stoichiometric formulas and X-ray diffraction information [6] on selected calcium and phosphate-containing phases

Name	Formula	JCPDS	Strongest powder XRD reflection (° 2θ)	*d*-spacing (Å)	Calcium to phosphorus molar ratio
Hydroxyapatite	$Ca_5(PO_4)_3OH$	09-0432	33.2	2.820	1.67
Calcium-deficient hydroxyapatite	$Ca_9(HPO_4)(PO_4)_5OH$	46-0905	33.2	2.820	1.50
Alpha-tricalcium phosphate	$Ca_3(PO_4)_2$	09-0348	32.0	2.910	1.50
Beta-tricalcium phosphate	$Ca_3(PO_4)_2$	09-0169	32.3	2.880	1.50
Dicalcium phosphate	$CaHPO_4$	09-0080	31.4	2.960	1.00

catalysts [9]. The general formula for these materials is [7]:

$$Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$$

The Ca/P of NSHA ranges from 1.67 to 1.33 as x varies from 0 to 2. When x is 1, the compound, $Ca_9(HPO_4)(PO_4)_5OH$, has a Ca/P molar ratio of 1.50 and is known as calcium-deficient hydroxyapatite [7]. Non-stoichiometric hydroxyapatite has the same crystal structure as hydroxyapatite [4].

The FTIR spectrum of commercial calcium phosphate adjuvant is identical to that reported for calcium-deficient hydroxyapatite [7] and is shown in Fig. 2. It is characterized by sharp, well-resolved bands at 1032, 603 and 563 cm^{-1}. The most intense feature in the IR spectrum is the 1032 cm^{-1} band with a shoulder at 1094 cm^{-1}. These bands correspond to P–O stretching vibrations. The broad bands at 3433, 3251 and 1640 cm^{-1} are due to the presence of both water and structural hydroxyl groups. The bands at 3565 and 630 cm^{-1} have been assigned to stretching and bending vibrations of the structural hydroxyl group [7]. The calcium phosphate adjuvant produced by Brenntag Biosector exhibited the same FTIR spectrum as shown in Fig. 2.

The TGA/DSC curves of the commercial calcium phosphate adjuvant are shown in Fig. 3. At temperatures <500 °C, the weight loss and shape of the TGA and DSC curves are due to dehydroxylation. The shape of the TGA curve shown in Fig. 3 is consistent with TGA of calcium deficient hydroxyapatite [10].

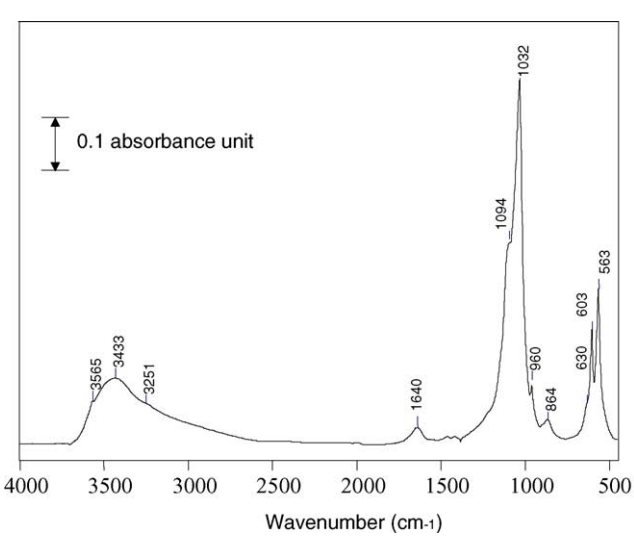

Fig. 2. FTIR spectrum of commercial calcium phosphate adjuvant.

Therefore, based on the XRD, Ca/P, FTIR spectrum, and thermal analysis, we conclude that commercial calcium phosphate adjuvant is non-stoichiometric hydroxyapatite ($Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$). The adjuvant is not the chemical calcium phosphate, $Ca_3(PO_4)_2$, as the name implies. The presence of a hydroxyl group is important as compounds containing metallic hydroxyls exhibit a pH-dependent surface charge [11]. In addition, adsorption of phosphorylated antigens by ligand exchange with surface hydroxyls may also occur [12].

3.2. Adsorption properties

A transmission electron photomicrograph of commercial calcium phosphate adjuvant (Fig. 4) revealed a mixture of particles: needle-like particles having dimensions of approximately 10 nm × 150 nm as well as irregularly shaped plates having diameters of approximately 20–30 nm. Calcium-deficient hydroxyapatite has been reported to consist of elongated hexagonal plates, regular parallelogram shaped plates and needles [8]. Thus, the morphology of calcium phosphate adjuvant observed in Fig. 4 is consistent with calcium-deficient hydroxyapatite.

The point of zero charge (PZC) of commercial calcium phosphate adjuvant was 5.5 (Fig. 5). The PZC of calcium-deficient hydroxyapatite was reported to range from 4.0 to 7.2 depending on the precipitation conditions [13]. The pH-dependent surface charge suggests that calcium phosphate

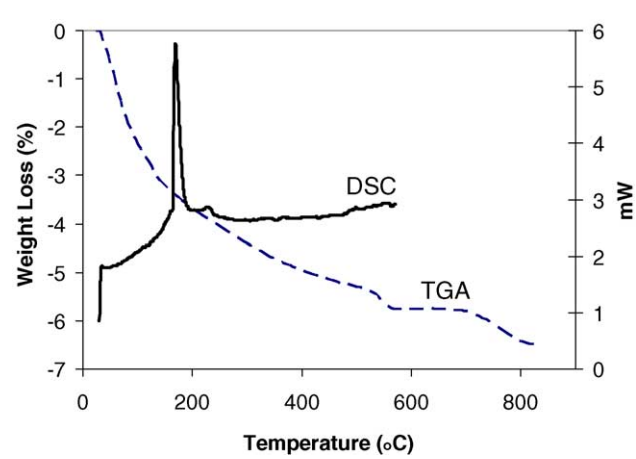

Fig. 3. Thermal analysis of commercial calcium phosphate adjuvant. Key: left ordinate, thermal gravimetric analysis; right ordinate, differential scanning calorimetry.

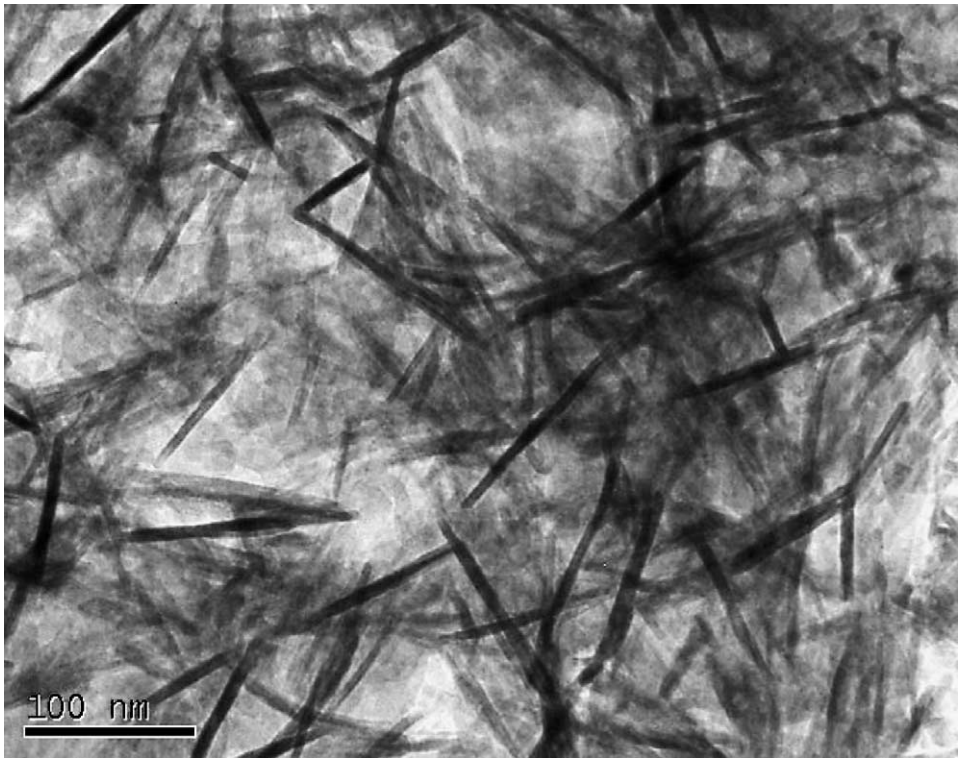

Fig. 4. Transmission electron photomicrograph of commercial calcium phosphate adjuvant. The bar represents 100 nm.

adjuvant will adsorb positively charged antigens at pH 7.4 by electrostatic attraction. The adsorption isotherm of lysozyme (iep = 11.0) at pH 7.4 in water is shown in Fig. 6. The adsorptive capacity was obtained from the reciprocal of the slope from the linear form of the Langmuir equation [14]. The adsorptive capacity was 0.13 mg lysozyme/mg Ca and is presented in Table 2.

The electrostatic adsorption of lysozyme by calcium phosphate adjuvant was tested by measuring the adsorption at pH 5.0. At this pH value, both lysozyme and calcium phosphate adjuvant were positively charged. No adsorption of lysozyme was detected (Table 2) even though the calcium phosphate adjuvant was exposed to lysozyme concentrations ranging from 0.1 to 1.0 mg/mL.

The electrostatic adsorption of lysozyme by calcium phosphate adjuvant was confirmed by determining the effect of ionic strength on the adsorptive capacity. As seen in Fig. 6, the adsorption of lysozyme at pH 7.4 was very sensitive to the ionic strength. The adsorptive capacity (Table 2) was zero in the presence of 0.05 M NaCl, which is less than the isotonic sodium chloride concentration of 0.15 M.

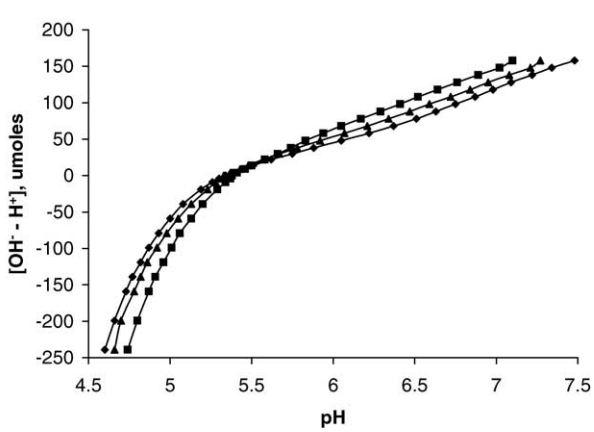

Fig. 5. Point of zero charge of commercial calcium phosphate adjuvant determined by potentiometric titration. Key: (♦) control; (▲) 0.2 mL of 4 M KCl added; (■) 1.0 mL of 4 M KCl added.

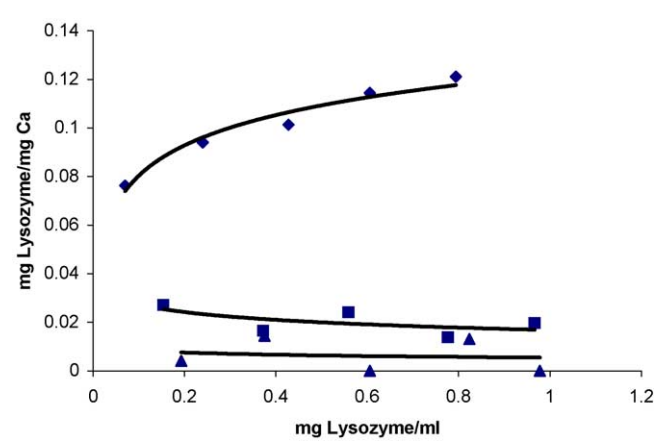

Fig. 6. Adsorption isotherms of lysozyme by commercial calcium phosphate adjuvant. Key: (♦) water; (■) 0.015 M NaCl; (▲) 0.05 M NaCl.

Table 2
Adsorptive capacity of commercial calcium phosphate adjuvant for lysozyme and alpha casein

Protein	Medium	pH	Adsorptive capacity (mg/mg Ca)	95% confidence interval (mg/mg Ca)	R^2
Lysozyme	Water	7.4	0.13	0.11–0.17	0.9903
	Water	5.0	0.00	–	–
	0.015 M NaCl	7.4	0.02	0.01–0.07	0.8499
	0.050 M NaCl	7.4	0.00	–	–
	20% EG[a]	7.4	0.13	0.11–0.14	0.9918
	40% EG[a]	7.4	0.11	0.10–0.13	0.9943
Alpha casein	Water	7.4	0.98	0.91–1.07	0.9963
	0.050 M NaCl	7.4	1.07	1.073–1.074	0.9999
	0.075 M NaCl	7.4	1.17	1.16–1.18	0.9999
	20% EG[a]	7.4	1.06	0.96–1.17	0.9952

[a] Ethylene glycol.

The potential of commercial calcium phosphate adjuvant to adsorb antigens by hydrophobic attractive forces was tested by determining the effect of ethylene glycol on the adsorption of lysozyme at pH 7.4. Ethylene glycol stabilizes the hydration layer of proteins, which renders hydrophobic interactions thermodynamically unstable [15]. Thus, a decrease in the adsorptive capacity in the presence of ethylene glycol is evidence of hydrophobic adsorption. As seen in Table 2, the addition of up to 40% ethylene glycol had no statistically significant effect on the adsorptive capacity. Thus, calcium phosphate adjuvant has little potential for hydrophobic adsorption of antigens.

The hydroxyl group in commercial calcium phosphate adjuvant makes it a candidate to strongly adsorb phosphorylated antigens as the adsorption of phosphate anion by hydroxylated mineral surfaces occurs by ligand exchange [12]. Previous studies on aluminum hydroxide adjuvant demonstrated that phosphorylated proteins and antigens are adsorbed by ligand exchange [16,17].

Alpha casein was selected as a model protein that has the potential to be adsorbed by ligand exchange as it contains eight phosphate groups [16]. Adsorption of alpha casein by calcium phosphate adjuvant in water at pH 7.4 followed the Langmuir equation (Table 2). The adsorptive capacity was 0.98 mg alpha casein/mg Ca. The isoelectric point of alpha casein is 4.6 [18]. Thus, the observed adsorption was not due to electrostatic attraction as both calcium phosphate adjuvant and alpha casein were negatively charged at pH 7.4. This behavior suggests that ligand exchange was responsible for the observed adsorption.

As seen in Table 2, the adsorptive capacity for alpha casein was not diminished by increased ionic strength. This observation further supports the conclusion that calcium phosphate adjuvant can adsorb phosphorylated antigens by ligand exchange.

There was no evidence that commercial calcium phosphate adjuvant adsorbed alpha casein by hydrophobic attractive forces as the adsorptive capacity was not affected by the presence of 20% ethylene glycol (Table 2). Higher concentrations of ethylene glycol could not be tested as alpha casein's solubility was reduced by ethylene glycol.

4. Conclusions

Commercial calcium phosphate adjuvant is chemically non-stoichiometric hydroxyapatite, $Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$. It is present as needle-like particles having dimensions of approximately 10 nm × 150 nm as well as irregularly shaped plates having diameters of approximately 20–30 nm. It has a pH dependent surface charge (point of zero charge 5.5) and adsorbs antigens by electrostatic attraction. It also adsorbs phosphorylated antigens by ligand exchange. It exhibited very little potential for hydrophobic adsorption of antigens.

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